Participatory Mapping and Air Quality Monitoring *for Cleaner Air in Nigeria*

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SUMMARY

In urban areas like Kaduna, Lagos, and Ogun in Nigeria, communities face increasing exposure to the health and climate risks of industrial air pollution—often with limited access to localized data. While government and academic institutions had started air quality monitoring, they faced barriers in spatial coverage and data accessibility. The Eco-Smart Cities - Nigeria Project, led by HOT's West and Northern Africa Open Mapping Hub, enhanced these efforts by scaling air quality monitoring to more areas. By combining participatory mapping and a blend of satellite and sensor-based air quality tracking, the project expanded data coverage, improved accuracy through localized monitoring, and strengthened collaboration across sectors—laying the foundation for better-informed planning to reduce pollution and protect public health.

UNDERSTANDING THE *CONTEXT*

Urbanization and industrialization have brought both progress and pressure to Nigeria: while they drive economic growth, they also result in increasing air pollution.

Research shows that back in 2019, there were 198,000 reported premature deaths that were attributable to air pollution in Nigeria (Clean Air Fund, n.d.).

In growing cities like Kaduna, Lagos, and Ogun, these dynamics have led to worsening air quality and mounting public health risks. The World Health Organization (n.d.) reports:

Engagement with Kaduna State University representatives (HePD, 2024).

"many of the sources of outdoor air pollution [such as industries and transport] are also sources of high CO2 emissions. [...] Pollutants not only severely impact public health, but also the Earth's climate."

1 All research, analysis, and conclusions are the authors' own. OpenAI was used to assist with drafting, and the text was reviewed, edited and validated by humans.

Air Quality Sensor Device used during data collection (HePD, 2024).

Before this project, local actors had laid an important foundation to address this issue. The Lagos State Environmental Protection Agency (LASEPA) and other environmental regulatory agencies operated air monitoring stations, while universities and NGOs carried out research and advocacy efforts.

However, air quality data remained fragmented and coverage limited—particularly in industrial zones. In Kaduna, Kaduna State University (KASU) reports that they lacked air quality sensors on the ground for their air quality research, leading them to be dependent on satellites using global modelling.

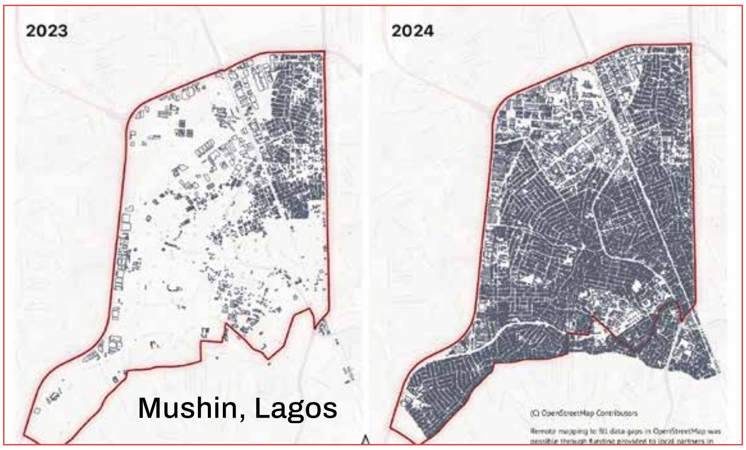
In Lagos, LASEPA had established citywide air quality monitoring, but also lacked ground air quality sensors for the few locations, creating what are known as 'dark spots' in their ground monitoring capacity.

The Eco-Smart Cities Project in Nigeria was designed to build on this foundation by increasing access to reliable, localized environmental data, and strengthening local capacity and collaboration towards cleaner air.

OUR APPROACH *IN ACTION*

In 2024, HOT's West and Northern Africa Open Mapping Hub supported the local youth-mapping organization Humanitarian enhanced Platform for Development (HePD) to team up with the Lagos, Kaduna, and Ogun States' environmental protection agencies (LASEPA, KEPA, and OGEPA, respectively) and universities (LASU, KASU, and OOU—Olabisi Onabanjo University) in a shared effort to improve how air quality is monitored and used for planning. Together, they followed a three-phase process: **Preparatory, Data Collection & Analysis, and Knowledge Sharing** (Humanitarian enhanced Platform for Development [HePD], 2024):

- **Preparatory:** the team reviewed existing studies to identify data gaps and trained local partners for project implementation.
- **Data Collection & Analysis:** the team led two efforts: (1) open mapping with student volunteers to fill critical gaps in OSM building and road data, and (2) air quality monitoring with low-cost, solar-powered sensors to measure key pollutants like nitrogen dioxide (NO_2), sullfur dioxide (SO_2), and particulate matters (PM_{10}, and $PM_{2.5}$) around industrial zones. This was made possible through collaboration with the state agencies and universities. They also received a total of six solar-powered ground air quality sensors, and were trained on how to use them.
- **Knowledge Sharing:** maps and initial findings were shared with stakeholders to support public engagement and local planning discussions.



Before and After mapping of Mushin, Lagos show increase of building footprints mapped in OSM.

Altogether, **over three million buildings, 6,000 km² of roads, and 38,895 km² of land were mapped in 22 local government areas** in Kaduna, Lagos, and Ogun (HePD, 2024).

OUTCOMES AND POTENTIAL IMPACT *ON PEOPLE AND COMMUNITIES*

Through this project, government agencies, academic institutions, and civil society groups gained access to open mapping platforms and solar-powered air quality sensors—enabling more community-driven and real-time data collection across target areas.

A preliminary air quality analysis in Kaduna surfaced that “the combined presence of these pollutants [NO_2 and SO_2] highlights the compounded health risks for residents exposed to this mixture of contaminants during peak traffic hours or periods of industrial activity” and a “moderate range” in Mushin, Lagos.

The analysis noted that “while air quality in Mushin was not severely hazardous,it remained a concern, particularly for vulnerable populations” (HOT’s West and Northern Africa Open Mapping Hub, 2025).[2]

This improved access allowed for the generation of localized insights into pollution levels—particularly in industrial zones where residents can be exposed to a mixture of contaminants. The initial findings underscored potential health risks especially to vulnerable groups like children and the elderly.

The participatory approach and initial findings of this project helped communities better understand local air pollution and its health impacts.

“The solution provided is unique and purposeful compared to the satellite estimates that do not [accurately] reflect the local situation. Running a 24-hour nonstop air quality monitoring station without any overhead cost was a game changer.”

Kaduna State University

HePD noted that “community awareness of air pollution issues has significantly improved,” (Humanitarian enhanced Platform for Development (HePD), personal communication, May 16, 2025) as people began to recognize pollution sources and risks in their daily lives.

Citizen group UrbanBetter, shared that this access to evidence is “enabling us to engage from an evidence-based perspective and to demand climate actions and responsibilities from our city administrators.” (UrbanBetter, personal communication, May 4, 2025). Meanwhile, government agencies are now better equipped to track air quality and take informed steps to protect over 3.5 million people in Nigeria.

In Lagos, LASEPA continues monitoring at newly established monitoring sites.

> **“With the addition of two solar-powered air quality monitoring sites, some of the ‘dark spots’ have now been illuminated with 24-hour observation. With complementary datasets [*from OSM*], we are better positioned to design emergency response plans for sister agencies in the event of a gaseous accident.”**
>
> Lagos State Environmental Protection Agency

Air Quality Sensor Device used during data collection (HePD, 2024).

While in Kaduna, Prof. Mande K. Hosea, project coordinator and Director of the Center for Climate Change and GIS at KASU, explained that sustained efforts by the university aim to produce analysis and recommendations by 2026 to support regulatory planning and shape air quality policies—**potentially benefiting over 2 million Kaduna residents.**

[2] Richard Djarbeng, the air quality expert consulted for this study, notes that the short timeframe of the on-ground air quality monitoring may not have captured the full pollution patterns in these areas. There is need for longer-term monitoring to better inform analysis and recommendations.

HOW CHANGE *CAN LAST*

This project strengthened sustainability by embedding tools, practices, and parnerships into local institutions.

The participating universities and government agencies have integrated the six solar-powered air quality sensors into their ongoing monitoring programs to support teaching, research, and regulatory work.

The collaborative, participatory approach has also been institutionalized by key actors.

Respondents from LASEPA, KASU, and UrbanBetter confirmed that they are continuing with monitoring efforts to inform future analysis and planning.

With guidance from air quality expert Richard Djarbeng, HOT will further develop a Standard Operating Procedure (SOP) to help local communities replicate this monitoring model in their own contexts.

INSIGHTS FOR THE *FUTURE*

The Eco-Smart Cities project in Nigeria demonstrated how open mapping and air quality monitoring can strengthen local capacities to address the climate and health challenges of urban air pollution.

By building on existing government efforts and fostering collaboration across sectors, the project provided tools, data, and approaches that supported local actors to generate early insights, engage communities, and begin embedding environmental data into their institutional processes.

Although full-scale policy change may take time, the foundation has been laid for more data-driven decision-making and participatory governance. With continued investment in collaborative monitoring and cross-sector dialogue, cities like Lagos, Kaduna, and Ogun are better positioned to advance air quality action and build more climate-resilient, health-conscious urban environments.

Mapping activity with the HePD team (HePD, 2024).

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Installation of Air Quality Monitoring equipment at the LASEPA office, Lagos (HePD, 2024).